

S Y M P O S I U M

WHAT ARE MICROFOUNDATIONS?

JAY BARNEY
University of Utah

TEPPO FELIN
Brigham Young University

In the extant organizational, management, and strategy literatures there are now frequent calls for microfoundations. However, there is little consensus on what microfoundations are and what they are not. In this paper we first (briefly) review the history of the microfoundations discussion and then discuss what microfoundations are and are not. We highlight four misconceptions or “half-truths” about microfoundations: (1) that microfoundations are psychology, human resources, or micro-organizational behavior, (2) that borrowed concepts constitute microfoundations, (3) that microfoundations lead to an infinite regress, and (4) that microfoundations deny the role of structure and institutions. We discuss both the partial truths and the misconceptions associated with the above understandings of microfoundations, and we argue that questions of social aggregation and emergence need to be center stage in any discussion of microfoundations. We link our arguments about microfoundations and aggregation with closely related calls for new areas of research, such as “behavioral strategy” and the domain of multilevel human capital research. We discuss various forms of social aggregation and also highlight associated opportunities for future research, with a specific focus on the origins of capabilities and competitive advantage.

The question of how to link micro factors with the macro continues to be a topic of debate and interest to social scientists (e.g., Harper & Lewis, 2012; Raub, Buskens, & Van Assen, 2011), including organizational, management, and strategy scholars (Felin, Foss, Heimeriks, & Madsen, 2012). In strategic management, for example, Coff and Kryscynski (2011) have called for identifying the microfoundations of competitive advantage (cf. Felin & Foss, 2005). In organization theory, Greve (2013) has called for understanding the microfoundations of organizational learning and behavioral strategy (cf. Gavetti, Greve, Levinthal, & Ocasio, 2012). And in the micro organizational behavior literature, Kozlowski and Chao (2012) have recently called for increased study of the microfoundations (and emergence) of collective-level constructs such as group cohesion and knowledge.¹

However, the sheer diversity of research that claims to be microfoundational in character suggests that there is little—if any—consensus about what actually constitutes microfoundations research. For example, Gavetti’s (2005) notion of microfoundations focuses on applying individual-level concepts, such as cognition and learning, to the firm level (cf. Gavetti, 2012). Eisenhardt, Furr, and Bingham (2010), by contrast, focus their discussion of microfoundations on the role of structure and simple rules in dynamic environments (cf. Eisenhardt & Martin, 2000). While both of the above papers focus on important issues, one can debate whether they indeed provide microfoundations. Neither article, for example, meaningfully addresses the central issue of social aggregation: micro-macro links—an issue that seemingly should be at the very core of any microfoundations discussion. Not only are there varied uses of and

¹ For several decades scholars have also called for organizational work to focus on the “meso” level linking

micro and macro organizational behavior (cf. House, Shane, & Herold, 1996; Mowday & Sutton, 1993; Rousseau, 1985).

connotations for the term *microfoundations*, some also argue that reductionist microfoundations are not even needed as the social sciences fundamentally deal with emergent phenomena and irreducible social complexity, and thus reduction and questions of aggregation simply are not central (e.g., Jepperson & Meyer, 2011; Winter, 2012a). In all, appeals (and challenges) to—as well as calls for—microfoundations are now ubiquitous, and thus we feel it is time to clarify and debate the precise nature of microfoundations in management and strategy. It seems unlikely that much progress will be made in this area of research until some consensus is reached about what microfoundational research actually is, and what it is not.

The purpose of this paper is to offer a clear definition of what does—and does not—constitute microfoundations research. The paper begins with a (brief) history of the microfoundations question (in the social sciences) and then discusses prevailing half-truths and misconceptions associated with the concept of microfoundations in organization theory, management, and strategy. We discuss the following half-truths: (1) that microfoundations are psychology, human resources (HR), or micro-organizational behavior (OB), (2) that borrowed concepts constitute microfoundations, (3) that microfoundations lead to an infinite regress, and (4) that microfoundations deny the role of structure and institutions. Thereafter we discuss what microfoundations are, specifically by highlighting both additive and more complex, “emergent” forms of aggregation, along with discussing associated implications for management and strategy. This discussion of microfoundations is also linked with recent calls to extend the behavioral theory of the firm into micro directions (e.g., Gavetti et al., 2012; Greve, 2013), as well as closely allied work that has called for a subfield of “behavioral strategy” (e.g., Levinthal, 2011; Powell, Lovaglio, & Fox, 2011), as well as recent calls for multilevel theories of human capital (e.g., Coff & Kryscynski, 2011; Molloy, Ployhart, & Wright, 2011; Ployhart & Moliterno, 2011). We conclude with a brief discussion of how our arguments about microfoundations and aggregation relate to the origins of organizational capabilities and competitive advantage.

MICRO OR MACROFOUNDATIONS: SOME BRIEF HISTORY ON THE DEBATE

Questions of the primacy of micro versus macrofoundations have been center stage in the social

sciences since the birth of the field. The earliest forms of this debate were focused on whether the social sciences should be based on methodological individualism or methodological collectivism (for an overview, see Udehn, 2001). Emile Durkheim (1962, p. 39, 106), often called the father of social science, advocated a collectivist methodology and argued that “social facts must be studied as things” and that “individual natures are merely the indeterminate material that the social factor molds and transforms.” The Durkheimian methodology dispensed with individuals—and the very notion of human nature—and argued that the social sciences, and sociology in particular, should focus on higher social and macro factors such as culture, religion, and even nation-states. The Durkheimian focus on the macro continues to be a point of emphasis by many in the social sciences (e.g., Jepperson & Meyer, 2011). It argues that the institutions, roles, rules, and structures of society are more important than individual-level factors in understanding society, markets, and individual behavior. This stream of research, then, privileges macro explanations and macro-to-macro relations, as well as macro-to-micro links, at the expense of individual nature, agency, and choice. The argument is that collective constructs such as institutions and structures are emergent and irreducible.

Durkheimian collectivism of course contrasts squarely with the methodological individualism advocated by many other social scientists, including such scholars as Georg Simmel and Max Weber. For these scholars the individual was the basic building block of social theories (cf. Simmel, 1974; Swedberg, 2007; Weber, 1949). In practice this meant that individuals’ beliefs, preferences, and interests provided a fruitful starting point from which to build theories of how social structures originate and evolve over time. This “methodological individualism”—although it admittedly can be construed in many strong and weak forms (cf. Udehn, 2002)—is the basic notion that to understand any collective phenomenon or thing, we need to understand the constituent parts that make it up: individuals and their social interaction. The idea is that collective givens need to be unpacked. Rather than simply postulate and point to the existence of collective things—whether structure, culture, institution, organization, market, or society—the precise job of the social scientist is to explain the

origins of the macro as the result of individuals and their interaction (Coleman, 1990).²

The debate between collectivists and individualists has also manifested itself in other disciplines, such as economics. For example, the *Methodenstreit* between the German Historical School of Economics (e.g., Schmoller) and the Austrian School (e.g., Menger, Hayek) was fundamentally a debate about whether explanation in the social sciences, and of the economy and markets in particular, should focus on collective wholes and history (advocated by the former) or on individuals and their interaction (advocated by the latter) (cf. Caldwell, 2005). Shades of this debate are mirrored in current debates about the microfoundations of organization theory and strategy: questions about the role of the environment and philosophical questions about the nature of economic actors (e.g., Felin & Foss, 2011; Hodgson, 2012; Hodgson & Knudsen, 2011a; Winter, 2011). A version of this debate has also existed in other areas such as economic anthropology—between formalists and substantivists (Hann & Hart, 2011)—where some scholars have argued for the importance of macro factors such as culture and history in determining the nature of and outcomes in markets and society (Polanyi, 1944), while others prefer to focus on the choices of actors and their aggregation (e.g., Barth, 2007).

It is worth noting that the social sciences have also been influenced by levels-related discussions in the physical sciences (cf. Oppenheim & Putnam, 1958; Schaffer, 2003). The British Emergentists, for example, had a layered, hierarchical view of nature in which lower levels give rise to higher levels. This intuition remains central in debates about how the brain, a physical thing, gives rise to higher-level nonphysical things, such as thought and consciousness (Kim, 1993). These discussions have also had an impact on the social sciences, specifically where areas such as the philosophy of mind have been directly applied to understand micro-macro links in the social sciences (see Sawyer, 2001).

The above historical sketch of microfoundations and associated debates (about individualism versus collectivism) is meant to highlight the interdis-

iplinary nature and origins of these discussions. Many of the same issues are at stake in organizational, management, and strategy research: the primacy of particular levels, the causal relations, and mechanisms across and between levels, among others. But notions of microfoundations are often caricatured. Some argue that microfoundations suggest an overly strong (and exclusive) focus on individuals. Others charge that microfoundations lead to an infinite regress. And yet others argue that microfoundations seem to deny the causal role of structure.

HALF-TRUTHS ABOUT MICROFOUNDATIONS

We next discuss some of the half-truths and misconceptions associated with microfoundations, particularly as these apply to the fields of management, organization, and strategy. These misconceptions are quite pervasive and thus deserve some attention and clarification. They represent widespread caricatures that are often voiced by those advocating a more macro approach, or by those who have a very specific conception of what microfoundations ought to be. Note, however, that these half-truths also contain a kernel of truth about what microfoundations in fact are (thus we label them “half”-truths), and we therefore discuss both the misconceptions and the partial truths associated with these understandings of microfoundations.

Half-Truth #1: Microfoundations Are Psychology, HR, or Micro-OB

The first misconception about microfoundations is that they are about individuals, and thus microfoundations are simply equivalent to more micro disciplines such as psychology, human resources, or micro-organizational behavior. That is, any macro concepts or theories should be reduced to explanations that focus on individuals: their characteristics, abilities, knowledge, cognition, or behaviors. Historically, some social scientists have indeed argued that all macro theories and concepts should be reduced to individuals and individual-level explanation (for an overview, see Udehn, 2002; cf. Elster, 1989). This approach essentially takes an additive perspective of collectives: Organizations and social systems are a function of the set of individuals contained within them. From a practical perspective, this means that we can understand organizations and their performance by carefully specifying the exact nature of these indi-

² Micro and macroeconomics have also been at odds with each other, particularly on the question of whether macro constructs, such as aggregate demand and supply curves, deserve more careful microfoundations (Janssen, 1993; Weintraub, 1979).

viduals—their personalities, abilities, and skills—and study organizational and social performance in simple, additive fashion. From a micro perspective, this mirrors the intuition of Schneider's (1987) attraction-selection-attribution model of organizational behavior, where organizations are a function of the individuals that constitute them.

But microfoundations are not solely about individuals. The problem with reducing everything to individuals, as we will discuss further below, is that it ignores the interactions among them as well as the context of the organization itself (cf. Whetten, Felin, & King, 2009). Individual interactions are not simply additive, but can take on complex forms and lead to surprising aggregate and emergent outcomes that are hard to predict based on knowledge of the constituent parts. Thus reducing, or attempting to reduce, everything to individuals is only “micro”—not *microfoundational*. In other words, the foundations portion of microfoundations is important in that it places emphasis on the need to specifically understand the unique, interactional, and collective effects that are not only additive but also emergent. “Adding individuals,” while important, leaves the hard work of actually aggregating up to the collective or organizational level undone. Therefore, microfoundations do not (solely) equal a focus on individuals.

That said, we might note that the “microfoundations as individuals (or psychology)” approach certainly has some virtues—thus suggesting that it is a “half”-truth—in terms of understanding organizational behavior and performance. This additive approach seeks to at least specify and understand the organization in terms of one of its central “inputs,” individuals, rather than postulate macro causes for individual behavior, thus jumping directly to macro factors such as culture or structure. In some situations, this approach, while perhaps limited, can in fact yield understanding about the nature of organizations and collective phenomena. To provide an example from economics, Hirschman's (1971) exit-voice-loyalty model of organizations is essentially an additive approach to understanding organizational behavior and performance. Also, areas such as social choice, game theory, and mechanism design in economics are additive, but they have nonetheless given us powerful intuition for understanding how preferences aggregate (Arrow, 1951; Sen, 1999). But these models, of course, do not take into account the interactional and relational factors that might influence preferences, col-

lective decision making, and aggregation (Stirling & Felin, 2013).

From a strategic perspective, the literature on talent and mobility is relevant and can also be seen as taking an additive perspective on organizations. It might be that in some organizations, in certain contexts, the performance of an organization can be relatively directly attributed to the talents of particular persons within the organization. Settings where there is little interdependence among individuals provide a good example. As discussed by Felin and Hesterly (2007), mobility indeed is a litmus test of whether we should focus on the individual or collective level.³ Scholars have begun to study how mobility affects organizational outcomes. For example, Aime, Johnson, and Ridge (2010) showed that team performance in the National Football League is not so much a function of routines or other collective constructs, but of the set of individuals who make up the organization. More specifically, if certain individuals leave, the performance of the whole organization suffers.

The issue of stars and individual performance of course raises tricky questions about value creation and rent appropriation (cf. Coff, 1999). Henderson and Cockburn (1994) in fact justified their macro approach, and assumptions about the homogeneity of human capital, by arguing that stars cannot be a source of sustainable advantage as they appropriate their marginal rents. In other words, the reason that some have focused directly on collective capability is that information about the capability of particular individuals (if markets are relatively efficient), such as stars, is likely to be widely available and thus these individuals are perhaps able to appropriate any rents associated with their abilities.

Overall, an additive approach has much to commend it in the sense that we know individual talents are not irrelevant to organizational performance, and talents and individual-level factors have important (and hard to impute) spillovers for the remainder of the organization. But talent can naturally also be a detriment in the sense that it

³ This contrasts with others who argue that we should focus strictly on the collective rather than the individual level: “If we truly focused on routines, competencies, practices, and so on, we would not follow people anymore in our research. Instead we would follow how competencies spread, replicate, and insinuate themselves into organizations. People would disappear from our equations” (Murmman, Aldrich, Levinthal, & Winter, 2003, p. 27).

might, in effect, disrupt value creation, appropriation, and collective processes (cf. Nickerson & Zenger, 2008). Thus we see that the aggregation of human capital is not simply additive, but careful attention needs to also be paid to aggregate, interactional, and emergent effects.

Half-Truth #2: Borrowed Concepts Constitute Microfoundations

Given the applied nature of the disciplines of management and strategy, there is a sense that the simple application of borrowed, individual-level concepts to the organizational level constitutes microfoundations. Concepts such as organizational learning, cognition, and organizational identity have been borrowed directly from individual-level antecedents and the associated literatures in psychology. Not just the concepts themselves are evoked, but the very same mechanisms are as well. The argument, roughly, is that since individuals learn (for example) through repetition, experience, association, and environmental feedback, then organizations must also do so.

Scholars have called this type of work microfoundational (e.g., Eisenhardt et al., 2010; Gavetti, 2005). For example, the organizational learning literature features the very same mechanisms (of repetition, environmental feedback, etc.) as the associated behavioral learning literature from psychology (Felin, 2012). Sometimes the concepts borrowed from other disciplines receive a different name when applied to the organizational level. For example, the concept of “absorptive capacity” (Cohen & Levinthal, 1990) is a concept directly borrowed from individual-level stimulus-response learning theories that focus on the importance of experience in learning and absorbing information (see Bower & Hilgard, 1981; Estes, 1970). The literature on cognition in strategy (e.g., Gavetti, Levinthal, & Rivkin, 2005; Gavetti, 2012; see also Gary, Wood, & Pilinger, 2012) also builds on individual-level concepts such as association and analogy (cf. Holyoak & Thagard, 1996). The typical pattern is to include a handful of citations to concepts in psychology, and then to invoke the concept and associated theoretical mechanisms at the organizational level.

We certainly think there are many virtues to borrowing and applying individual-level concepts to the organizational level. But even when these concepts are borrowed from more micro disciplines, such as psychology, the act of borrowing alone does not constitute a microfoundation simply be-

cause the concept originated from a more micro discipline. Not only are there questions and debates about these concepts and mechanisms in the disciplines from which they originate, there are of course also important questions about whether the nature of the borrowed concepts themselves (and associated theoretical mechanisms) needs to change when applied to the setting of organizations, management, or strategy. For example, biases exist in the individuals who constitute organizations, but the more important question is how the concept of bias applies to an organization, whether in the aggregate or as a unique social actor (cf. King, Felin, & Whetten, 2010).

It might be noted that some have argued for general or universal theories in which theoretical concepts and mechanisms, across levels and contexts, are the same. For example, the capabilities literature that builds on evolutionary economics is explicitly focused on general mechanisms such as environmental selection and a more general advocacy of universal Darwinism (Hodgson & Knudsen, 2011b; Winter, 2011). While this type of grand, universal theorizing is interesting, we think there is also a need for theories to clearly specify the nature of particular actors—whether individual or collective—rather than assuming that these universal theories readily apply across species, fields, and disciplines (Felin, 2012).

In all, while the borrowing of concepts across disciplines is natural for more applied fields such as management, organization theory, and strategy, there are nonetheless unique theoretical considerations that need to be carefully explicated when borrowing micro theories. Simply referencing a micro concept, such as learning or cognition, does not suffice and is not a microfoundation. One needs to have a meta-theory of how the concept itself needs to change or evolve given aggregation and interaction in the context of an organization or other social setting.

Half-Truth #3: Microfoundations Lead to an Infinite Regress

Another misconception about microfoundations is that they lead us to an infinite regress. That is, if organizational and social analysis needs to engage in reduction by looking at lower, micro levels—individuals and their interaction—then where should this reduction stop? Should organizational analysis be reduced beyond individuals to brains or genes—or perhaps parentage, ancestry, or even,

say, the big bang? This critique has recently been raised by Hodgson (2012) and Winter (2012a; also see Winter, 2013).

Infinite regress suggests a hierarchy of disciplines, from the social sciences to psychology, biology, chemistry, and physics (Oppenheim & Putnam, 1958), which raises the question of whether indeed there is infinite regress or some reasonable stopping point and “fundamental level” of analysis (cf. Schaffer, 2003). We might note that some scholars in management have in fact begun to look at the brain and its role in managerial settings (e.g., Laureiro-Martinez, Brusoni, & Zollo, 2010; Powell & Puccinelli, 2012; Volk & Kohler, 2012). In the sciences more broadly, there are efforts to “translate up” from lower-level neuroscience to higher levels of analysis (for an overview, see Levallois, Clithero, Wouters, Smidts, & Huettel, 2012). But if progress in the sciences, including the social sciences, indeed happens via reduction (Elster, 1989), is there a stopping point to this reduction? This infinite regress question of course also applies to microfoundations. To illustrate, if we say that an individual has a particular preference or skill or capability, the immediate question is “why?” Where did this individual-level capability come from? We might feasibly demand that the explanation of a capability extend further to one’s education or parents or even genes.

Microfoundations, however, do not necessarily demand extreme reduction. Specifically, the infinite regress problem can be “solved” in the sense that there are natural punctuations, initial conditions, and starting points for organizational and social analysis, and the individual provides a particularly salient one. With our focus on social aggregation as the central issue of microfoundations (see further discussion below), individuals then provide “a natural stopping point” for reduction and the appropriate starting point for analysis in the social sciences (Coleman, 1990, pp. 1–5). Rather than temporally reduce investigation of organizations into, say, the life histories of individual employees (cf. Beckman, 2006)—as valuable as that can be—we can take the individual as a given (his or her skills, preferences, and interests) and build models that focus on aggregation and the emergent outcomes of social interaction within organizations. As the levels of analysis literature points out, there are fundamentally emergent outcomes, interactions, and processes that deserve careful attention (Dansereau, Yammarino, & Kohles, 1999).

Some theories of course would question the very concept of the individual as the appropriate starting point for organizational analysis or the solution to the infinite regress problem. These theories, explicitly or implicitly, “question the notion of the individual” (e.g., Spender, 1996, pp. 52–54; see also Jepperson & Meyer, 2011; Kogut & Zander, 1992; Nelson & Winter, 1982, p. 63; Zollo & Winter, 2002), and thus our premise of starting with individuals and their interaction might be seen as provocative and not satisfy these scholars. But questioning the notion of individuals has led scholars to, in turn, make overly tenuous assumptions about individuals: that human capital is homogeneous, and that mobility and human nature simply do not matter in organizational analysis (for an overview, see Felin & Hesterly, 2007).

The focus on the “who”—the set of individuals who constitute an organization—then amounts to an effort to more carefully try to specify an initial condition for organizations. From the perspective of rigorous scientific explanation, as noted by Popper (1959, p. 198), specifying initial conditions is central: “If the initial conditions cannot be ascertained, the scientific way of predicting breaks down.” For organizations, these initial conditions do not just represent “who” is in the organization; they can also include temporal initial conditions, such as the founding of an organization (Felin & Knudsen, 2012). The choices that are made early in the history of an organization clearly have far-reaching consequences for the set of opportunities and advantages—or disadvantages, for that matter—that exist for an organization. Stinchcombe’s (1965) notion of “imprinting” captures this (cf. Aldrich, 1999), though the focus has generally been on how the environment, rather than the actors or their choices and interactions (cf. Johnson, 2007), leaves a long-lasting imprint on organizations and their path-dependent direction.

Now, the problem of infinite regress of course contains some truth. For example, any individual-level starting point (say, an individual’s capability or skill) might have temporal antecedents that extend far back into the life history of that individual: jobs that the employee may have had, connections and relationships he or she might have, and so on. Perhaps some causes of capability are temporally more “proximate” while others are more “ultimate” (cf. Mayr, 1961)—and both need to be understood—though there are of course questions about the primacy of the respective causes and their interrelationships. Furthermore, perhaps for some pur-

poses—depending on the dependent variable in question—it might indeed be useful to reduce further back to study how the history or experiences of particular individuals might lead to certain organizational outcomes.

But the fields of management, organization, and strategy are also uniquely independent disciplines with their own level of analysis: the organization (cf. King et al., 2010). While reducing explanations to genes or brain activity can be interesting and important, this work is not likely to fully inform the very practical, day-to-day activities and behaviors that managers and individuals in organizations engage in.

Half-Truth #4: Microfoundations Deny the Role of Structure and Institutions

Another half-truth about microfoundations is that they deny the role that structure—and other macro factors such as culture, institutions, and norms—should play in organizational and social analysis (e.g., Hodgson, 2012; Winter, 2013). This half-truth is related to the first half-truth we discussed, that microfoundations demand that everything be reduced to individuals. Scholars in sociology (Jepperson & Meyer, 2011), for example, have recently caricatured microfoundations by arguing that they miss the important influence of social-organizational processes, historical institutions, roles and cultural meanings, and other structures in society. They further argue that calls for microfoundations are “microchauvinist” and exclusionary of all macro factors. Similar arguments can be found in recent challenges to agentic theories of capability, arguing that more emphasis should instead be placed on serendipity and “contextual factors” (Winter, 2012b).

Those who have challenged the microfoundations literature somehow seem to suggest that microfoundations call for the ontological rejection of any higher-level units above individuals. This simply is not the case. The call for microfoundations is a *methodological* point about the power of looking at lower-level constituent units when explaining higher levels of analysis. In other words, the precise point of the microfoundations program is to systematically look at the *origins* and nature of the macro: how choices and interactions create structure, the behavior of individuals within structures, and the role of individuals in shaping the evolution of structures over time (cf. Chwe, 2001). Thus, rather than take structural and macro factors for

granted, the goal of the microfoundations program is to explain their origins and evolution by looking at how they emerge because of individual choices and social interaction. Thus, microfoundations in effect shift the causal arrow from macro-micro (or macro-macro) analysis to micro-macro analysis.

The problem with structural and situational explanations is that they often lead to the creation of black boxes rather than theoretical explanations. For example, Winter’s (2011) recent call for strategy scholars to focus on “contextual factors” in fact raises micro questions: Whose (objective) context are we talking about? Presumably, among different actors, there is large variance in these contextual factors. Or is there indeed some macro-environmental context or situation that we can speak about? And how do the heterogeneous capabilities and subjective perceptions of actors lead to some common context? Note that problematizing context and situation doesn’t require us to deny that there indeed are objective factors (such as competition, demand, and extant technologies). But the uncertainty of behavior in markets necessarily requires us to understand the subjective judgments and heterogeneous perceptions of actors that lead to these macro factors and their salience.

But, in all, there is of course also room within the microfoundations program of research to study and understand the role of structure in shaping individual-level outcomes, macro to micro causation. In other words, once the origins of structures have been explained, these structures in turn have causal influence on agents and their interaction. Environments, for example, provide an exogenous selection mechanism (cf. Aldrich, 1999). Naturally individuals and organizations also have choice within the constraints of their environments (Ingram & Clay, 2000). In other words, the microfoundations program is amenable to specifying macro-micro relationships, though it also recognizes that individuals and organizations have choices that create and shape the macro environment.

The problem with many macro approaches is that they abstract their analysis completely away from individuals to macro-macro type analysis. But we agree with Selznick that “no social process can be understood save it is located in the behavior of individuals” (1957, p. 4). Neo-institutional theories of organization, to give but one example, have placed a strong emphasis on various macro-macro relationships, and thus they emphasize “supraindividual units of analysis that cannot be reduced to aggregations or direct consequences of individuals’

attributes or motives" (DiMaggio & Powell, 1991, p. 8). But as further discussed by Selznick (1996), this macro-macro focus precludes us from actually understanding the role that individuals—their cognition, perceptions, and interaction—play in generating, sustaining, and changing institutions.

Microfoundations, then, are not a grand theory or dogma of any sort, but rather a pragmatic observation that explanation often is best accomplished by looking at the origins and evolution of collective givens as a function of lower-level factors and social interaction. Thus the truth portion of this "half"-truth is that a microfoundations program of research indeed has a preference for explaining the macro by focusing on the micro, rather than taking the former for granted. And what is further true about microfoundations is that macro-to-macro-type relationships are not allowed into the analysis (Abell, Felin, & Foss, 2008). That is, individuals and their interactions are central for understanding organizations and social systems.

Before proceeding, we might briefly note that there are of course ongoing efforts to move beyond any distinctions between micro and macro, such as Giddens' (1984) notion of "structuration." The argument here, essentially, is for a level-less ontology, where factors such as practice in essence represent the common ground or interface where the micro and macro mutually instantiate each other. This has led to a large and growing literature in areas such as strategy-as-practice (Whittington, 2003). The focus has been on such factors as the role of artifacts and boundary objects in social interaction and knowledge generation (Carlile, 2002). The strategy-as-practice area shares a broad affinity with the microfoundations program in that it seeks to more carefully delve into the actual micro activities, behaviors, and processes of strategy and organization (cf. Langley, 2007; Pettigrew, 1992). However, at a broad, philosophical level there are also some differences. The effort to remove the dichotomy between micro and macro, with concepts such as structuration (or habitus), is not completely satisfactory from a microfoundations perspective as the precise point of the microfoundations program is to try to more rigorously understand the constituent elements and interactions that make up a collective or system. Put differently, the problem with concepts such as structuration is that they, as argued by Archer (1995, p. 102), "[throw] a blanket over the two constituents, structure and agency, which only serves to prevent us from examining what is going on beneath it." That

said, we think there certainly are opportunities for future research to carefully unpack the intersection of microfoundations and programs of research that focus on areas such as strategy process and strategy-as-practice.⁴

MICROFOUNDATIONS AND AGGREGATION

While microfoundations can be construed in many ways, we argue that aggregation is the *sine qua non* of microfoundations, particularly in the domain of management, organization theory, and strategy. Thus, organizational analysis should be fundamentally concerned with how individual-level factors aggregate to the collective level. Early behavioral theories of organization in fact focused directly on this issue (e.g., March, 1962), though the focus on aggregation has subsequently been lost (as also noted by Gavetti, Levinthal, & Ocasio, 2007, p. 524; see also Knudsen & Levinthal, 2007, pp. 39–40). Scholars have instead placed their emphasis on populations of organizations and concepts such as organization-level learning, cognition, and environmental feedback. Thus, we next discuss central issues associated with social aggregation. We first discuss simple aggregation and the independence of individuals and then more complex forms of aggregation, emergence, and reciprocal influence.

Additive Aggregation and the Independence of Individuals

In some situations, aggregation is additive and simple. For example, some views of markets take this perspective. Actors are independent of each other, with their own preferences and interests, and various tools such as equilibrium-based analysis, game theory, and social choice are used to study aggregate outcomes and social interaction (Arrow, 1951). We certainly have no quarrel with this type of analysis, as it is quite powerful for helping us to understand a number of social phenomena.

Beyond markets and economics, many theories of organization also build on an additive view of collectives. For example, Schneider's (1987) afore-

⁴ Providing some promising links between microfoundations and practice-oriented research, scholars have recently studied how agency and intentionality interact with and shape the evolution of routines and behavior (Bapuji, Hora, & Saeed, 2012).

mentioned attraction-selection-attrition (ASA) model is precisely an effort to highlight how individuals self-select into organizations that fit their interests, capabilities, and skills (cf. Chatman, 1991; Harrison & Carroll, 2005). Schneider's model is a reaction to overly socialized models of organizational behavioral that give overpowering influence to the situation-related aspects of behavior at the expense of individuals (also known as the person-situation debate; see Davis-Blake & Pfeffer, 1989). The top management team literature, of course, can also be seen as one attempt to look at organizations as a type of aggregate. The focus is on the backgrounds, skills, experiences, and decision making of those at the top of an organization, and thus this literature sees the "organization as a reflection of its top managers" (e.g., Hambrick & Mason, 1984, p. 193).

More generally, notions such as homophily also provide an additive, aggregate perspective of organizations and collective phenomena (for an overview, see McPherson, Smith-Lovin, & Cook, 2001). Homophily is the tendency of similar individuals to self-select to interact with each other. This tendency applies for both value-related (e.g., interests, beliefs, expectations) and demographic (e.g., age, sex, race, education) characteristics. Thus, organizations could, in the aggregate, be of a certain variety based on similar individuals selecting into them. Homophily, then, assumes an additive, aggregate approach to collectives. The approach more specifically assumes that *ex ante* characteristics or values matter more than social influence and interactional processes. As Katz and Lazarsfeld (1955, pp. 59–60) put it, "common values precede rather than follow from social interaction." The notion of homophily has recently been applied by Felin and Knudsen (2012) to the setting of nascent organizations, entrepreneurship, and strategy.

The process of matching individuals and talent with organizations also provides a compelling, largely additive perspective on organizations and strategy (Felin & Zenger, 2011; see also Argyres, Felin, Foss, & Zenger, 2012). This research focuses on how the sorting, self-selection, and mobility of individuals—with heterogeneous capabilities—impact organizational boundaries and organizational performance. A roughly similar, additive perspective on organizations is also evident in the literature on strategic human capital. Scholars have recently called for additional work in this area, the nexus of strategy and human resources (cf. Campbell, Coff, & Kryscynski, 2010; Coff & Kryscynski,

2011; Moliterno & Ployhart, 2011). Work on talent and human capital indeed provides a promising way to study microfoundations in organization theory, management, and strategy. This research of course also raises important questions about the appropriate level of analysis for studying organizational advantage (Coff, 1999). Beyond an additive notion, human capital scholars have also argued that human capital is a complex, multilevel concept (Moliterno & Ployhart, 2011), including not only such individual-level factors as knowledge, skills, and abilities but also a host of social factors such as social capital and organizational culture. Teasing out the respective effects of each of these factors seems important. One might think about this from the perspective of variance decomposition: At what level of analysis does most of the performance-related variance exist? Simply referencing the need for "multilevel" theories is not sufficient. Rather, we need comparative theories and associated empirical analysis that prioritizes different levels of analysis in terms of their respective contribution to overall performance.

Recent calls for "behavioral" research also provide an opportunity to study questions of aggregation. The behavioral program of research encompasses both the call for strategy to link with behavioral economics and behavioral finance (e.g., Powell et al., 2011) and calls to extend the behavioral theory of the firm into new directions (e.g., Gavetti et al., 2012; Greve, 2013). For example, Powell et al.'s (2011, p. 1374) question "how does individual cognition scale to collective behavior?" represents precisely the type of work we think is needed in the organizational and strategy domains. Related to this, Foss and Lindenberg (2013) in fact developed a theory that seeks to link cognition with organizational goals and framing. But while there is a clear need to draw insights from psychology, economics, and other neighboring disciplines, the question of how behavior "scales" or aggregates is a central one—scarcely addressed in the extant organizational literature. For example, Simon's (1956) notion of bounded rationality has been exhaustively studied at the individual and firm level, but we know little about how bounded rationalities aggregate. This aggregation process, of course, may have linear and additive elements—though, the aggregation of bounded rationalities is likely to also feature surprising, emergent outcomes at the collective level. It is in areas such as the aggregation of bounded rationalities that we start seeing some of

the limitations of additive aggregation. Thus we next discuss more complex forms of aggregation.

Complex Aggregation, “Emergence,” and (Reciprocal) Influence

While simple, additive aggregation is helpful for understanding some organizational and social phenomena, aggregation in many settings is far more complex due to social interaction, interdependencies, and influence. In other words, individuals influence each other, and their interaction may lead to aggregate outcomes that can be unforeseen, surprising, and emergent. Social interaction can take many forms, leading to both positive (whole is greater than the sum of the parts) and negative (whole is less than the sum of the parts) effects.⁵ There is a general bias toward the positive aspects of social interaction (often labeled “synergies”), though the pathologies of social interaction and associated emergent outcomes are also important to specify and understand.

Before we proceed, the notion of “emergence” deserves some brief attention. Emergence is frequently appealed to in the social sciences (for an overview, see Sawyer, 2005). But the problem is that appealing to emergence often obfuscates explanation by hiding the actual mechanisms, processes, and actors that lead to the emergent outcome. Jepsen and Meyer (2011), for example, appeal to emergence and social complexity as reasons *not* to focus on microfoundations. But, as noted by Nagel (1961, p. 371), appealing to or labeling something as emergent only “baptizes our ignorance” rather than actually offering a scientific explanation of how the emergent outcome came about. The levels of analysis literature has also struggled with the notion of emergence (Dansereau et al., 1999). But emergence is not *ex nihilo* creation, something from nothing. Rather, the underlying constituent elements, the individuals interacting, need to stay in sharp focus when explaining aggregate, including emergent, social phenomena. And importantly, the fact that something might be irreducible does not mean that we cannot postulate a theory of how this irreducibility emerges because of agent interaction. Thus, it is not helpful to simply point to emergence or social complexity, or to simply

label something as emergent; rather, more explicit mechanisms are needed.

Our use of the word emergent, then, is more directly anchored on the notion that the nature of both the constituent individuals and their social interaction need to be carefully specified, recognizing that this interaction can also lead to collective outcomes that are surprising and not necessarily reducible to the constituent individuals. For example, Thomas Schelling’s (1971, 1978) simple and powerful model of segregation illustrates this. Schelling showed how the choices of individuals, who have only a slight preference for living next to someone of their own race, can lead to unanticipated, full-scale segregation at the macro level. This model, of course, could be further complicated by adding individual-level heterogeneity or even influence dynamics. But the more general point here is that individual interaction can lead to surprising and unintended macro-level outcomes once the emergent interaction is worked out to the macro level. No one agent is desirous of full-scale segregation, but the outcome nonetheless emerges due to the interaction of the agents. Agent-based simulation in the social sciences, indeed, provides a good example and opportunity to study complex social aggregation (for an overview, see Macy & Willer, 2002; also see Cederman, 2005).

While markets are often treated in additive, atomistic fashion in which individuals are independent of each other, markets of course also manifest complex forms of aggregation and often surprising, emergent outcomes (cf. Hayek, 1945). Thus the independence of market actors can be a tenuous assumption (cf. Fehr & Tyran, 2005). For example, we have manifestations of various market phenomena, such as bubbles, that are the result of complex aggregation, where agents reciprocally influence each other’s beliefs about the nature of reality or the value of assets and strategies—leading to surprisingly macro outcomes. Bank runs can be modeled based on rumors spreading false beliefs about the viability of a bank, leading to the collapse of the bank (Banerjee, 1992; Vaugiraud, 2005). Tulip mania and other bubbles also are manifestations of this, and have led scholars to begin looking at the behavioral and social foundations of markets (cf. Porter & Smith, 2003; Shiller, 2003).

But beyond markets, other social settings are also characterized by complex aggregation and emergent, collective outcomes. Much of the early findings in social psychology focused on social influence and associated dynamics (for an overview, see

⁵ There is also an opportunity to build on literatures in philosophy (specifically the area of mereology) that deal explicitly with part-whole relations (see Schaffer, 2003).

Stangor, 2009). For example, Solomon Asch (1951; cf. Sherif & Hovland, 1961), in his famous line experiments, found that individuals distort their judgments of reality even when only passively influenced by others. Other surprising or aberrant macro outcomes of social interaction include cognitive and social loafing, groupthink, self-censoring, diffusion of responsibility, deindividuation, and group inhibition. In short, social interaction can lead to surprising group-level pathologies that are hard to predict with simple knowledge of the constituent individuals.

Comparative work on brainstorming aptly illustrates aggregation in situations where individuals are independent versus situations where individuals interact. This research has compared the productivity of independent members of a group ("nominal" groups) to that of interdependent members (interacting or "real" groups), and the research indeed finds that independent individuals come up with a far larger and heterogeneous set of ideas than interacting individuals (Diehl & Stroebe, 1991; Rietzschel, Nijstad, & Stroebe, 2006). Interacting groups are far less productive due to the pathologies listed above (e.g., groupthink). This research, then, highlights how the macro fact of (simply) being in an interacting group influences individual judgments, perception, and productivity. Aggregate outcomes, then, can vary significantly depending on the (allowed or naturally occurring) interactions of the constituent parts.

Of course, there is a large body of work on the positive aspects of social interaction. For example, team or group cohesion is increased via social interaction—as is mutual knowledge and learning (see Kozlowski & Chao, 2012). The literature on "transactive memory" also shows how individuals both learn from each other and learn who has the relevant knowledge within the organization (Argote & Ren, 2012; Lewis & Herndon, 2011). Interaction, then, can of course have positive aggregate effects as individuals learn from each other—though long-term interaction and socialization can also lead to collective myopia (March, 1991).

While the teams and groups research (and social psychology more generally) has helped us learn about aggregation, for organizational and strategy scholars the problem of aggregation of course is far more complex, given our interest in much larger aggregates: organizations (as well as dependent variables such as competitive advantage). That is, individuals aggregate to teams, to groups of teams, and to organizations in nonlinear ways. The com-

plexities of this aggregation are an order of magnitude more difficult. The difficulty of this larger-scale aggregation has led organizational and strategy scholars to use shortcuts such as treating the organization as a unitary actor and appealing to "emergent" (but unexplained) collective effects, such as organizational capability or routines. And yet others have bypassed the problem of aggregation altogether by focusing on a handful of individuals within organizations, such as the top management team (and their traits or experience). While firm-level research certainly has been central in helping us understand questions such as the origins of competitive advantage, a natural next step is to "unpack" these aggregates to understand how organizational factors and advantages emerge.

MICROFOUNDATIONS OF CAPABILITY: FUTURE DIRECTIONS

We next discuss the implications of our arguments on questions of the origins of capability and competitive advantage, concurrently highlighting opportunities for future research.

The origins of competitive advantage can be traced to the unique information or expectations of a firm in a market (Barney, 1986) or to serendipity and luck (Alchian, 1950; Denrell, Fang, & Winter, 2003; Winter, 2013). In a "factor market," some resources might, in essence, be underpriced due to the unique knowledge or information possessed by a firm. The focus on firms as possessors of unique knowledge or information of course is shorthand in the sense that firms are collections of individuals. Information and knowledge are not possessed by the firm (cf. Nelson & Winter, 1982) *per se*, but rather by the individuals within it (though, for an overview of this debate, see Felin & Hesterly, 2007). Or the joint knowledge of many individuals is (somehow) aggregated up to firm-level knowledge. Individuals within an organization might also have wide-ranging, even conflicting, information and expectations about the most fruitful course(s) of action.

However, theories of strategy, such as factor markets (Barney, 1986), as well as theories of the firm (cf. Malmgren, 1961) have generally focused on the information and expectations of *singular* actors: firms or singular "entrepreneur-coordinators" who—for *convenience of the theory* (and not because it necessarily represents reality)—are said to make decisions on behalf of the organization (e.g., Coase, 1937). While this firm-level theorizing has

led to many central insights, microfoundations are also needed. In particular, we think that further understanding organizational capability and heterogeneity ought to rest on questions of microfoundations: how capabilities are built, how the matching of individuals with organizations occurs, the role of specific actors in building capability, and other, more general questions related to aggregation. Furthermore, there are related questions about the decision-making associated with building capabilities and the emergence of heterogeneity in markets.

The concern, then, is with various collective “givens” that need to be unpacked to understand factors such as organizational capability and performance. As March noted decades ago: “The composition of the firm is not given; it is negotiated. The goals of the firm are not given; they are bargained” (1962, p. 672). The same can be said for preferences, information, and expectations. These matters have not been addressed systematically in the literature on organizational capabilities, or the literature on competitive advantage. The hard work of aggregation needs to occur with regard to our extant theories of strategy and the theory of the firm (cf. Felin & Zenger, 2011). The first-order questions, then, should focus on the composition of the organization: Who—with what skills, abilities, capability, and knowledge—selects into (or leaves) organizations, with what aggregate effects? How is capability, in the aggregate, built? Where do firm-level expectations and information come from? How is information aggregated?

What is perhaps most directly relevant to organizational and strategy scholars, specifically those interested in capabilities, are questions of organizational design. Organizational design and structure play a central role in how information is aggregated (Stinchcombe, 1990). Design and structure is a way of purposefully delineating who interacts and communicates with whom, who has ultimate decision rights over what, and so forth. The capabilities of individuals, and thus organizations, may remain dormant or latent if organizations are poorly designed (Felin, 2012).

We do, for example, know that certain designs or organizational forms—such as polyarchy—can have beneficial outcomes for organizational decision making compared to other forms, such as hierarchy (Sah & Stiglitz, 1986; cf. Knudsen & Levinthal, 2007). The architectures of human and social interaction are central for determining the aggregate outcomes and collective capabilities we might observe. These architectures can constrain or

enable collective action. Of course, not all aggregation in organizations is “planned” or designed; organizations may also rely on more market-like, spontaneous order (see Foss, 2003).

Recent literature in the domain of innovation has also wrestled with questions about the aggregation of information and knowledge in organizations (e.g., Afuah & Tucci, 2012; Jeppesen & Lakhani, 2010; Nickerson & Zenger, 2004). The central question raised by this research is essentially both a microfoundational and aggregational one: How do we identify who has the correct information? Or, if information is dispersed, how do we aggregate it? Firms are striving to tap into the market’s proverbial “wisdom of crowds” via various practices such as crowdsourcing and prediction markets (cf. Felin & Zenger, 2011). Many questions remain about how factors such as incentives affect the identification of appropriate sources of knowledge and the aggregation of dispersed information and the evolution of capabilities.

As discussed previously, the fields of strategy and organization theory should specifically be concerned with the organizational level, and thus these fields also need to address the organization itself—beyond the individuals in them, and beyond their environments. In other words, beyond aggregation, how do we properly specify the organizational factors associated with capability development and performance? Instead of merely ascribing individual-level attributes to organizations or simply borrowing individual-level theories and applying them to organizations—both quite common in extant work—we need to move to the next step of actually articulating what makes capability development in organizations unique. This, again, does not excuse us to ignore individuals in organizations; rather, it requires us to understand how the organization itself, as a social context, affects and shapes individual behavior and (both individual and organizational) performance. Again, it is not sufficient to simply say that because individuals suffer from biases, so do organizations, or that because individuals learn, organizations learn. Rather, we need theories of organization and strategy that properly deal with the fact that aggregate capability development is happening within an organization. There are very few theories that explicitly deal with both the aggregate and contextual factors in organizations. As articulated by Heath and Sitkin (2001), we need theories not just of behavior, but of behavior both *in* and *of* organizations. Shades of this issue were foreshadowed by

Art Stinchcombe, who argued that any theory of organization must “explain how organizations can be more rational than individuals (though of course they are not always)” (1990, p. 341).

In all, we think there are significant opportunities for management, strategy, and organizational scholars to begin to address microfoundations: micro-macro links and social aggregation in organizations. This means that organizational work, beyond applying theories and insights from other disciplines, needs proper theories of aggregation, unique to organizations. We think that comparative insights about aggregation in fact might be drawn from other disciplines (such as physics, biology, and economics), but organizational scholars also need to engage in the hard work of specifying unique theories of aggregation that appropriately represent the social interactional and contextual factors that shape behavior and performance in organizations.

CONCLUSIONS

The purpose of this paper has been to discuss microfoundations in management and strategy—what they are, and what they are not. There are continuous, near-ubiquitous calls for microfoundations but little agreement about what they in fact are. We have outlined several half-truths and misconceptions propagated about microfoundations: that they (1) imply a focus solely on individuals, (2) lead to an infinite regress, (3) can be dealt with by borrowing theories, and (4) deny the role of structure. We also discuss simple, additive forms of aggregation—suitable for some purposes of understanding organizations and strategy—and more complex, nonlinear, and “emergent” forms of aggregation. Our hope is that this paper offers some clarity on what microfoundations are and what they are not—or at least generates further debate on the precise nature of microfoundations. Further, we hope that this paper triggers new work on central questions associated with social aggregation and emergence.

REFERENCES

- Abell, P., Felin, T., & Foss, N. (2008). Building microfoundations for the routines, capabilities, and performance links. *Managerial and Decision Economics*, 29, 489–502.
- Afuah, A., & Tucci, C. I. (2012). Crowdsourcing as a solution to distant search. *Academy of Management Review*, 27, 355–375.
- Aime, F., Johnson, S., & Ridge, J. W. (2010). The routine may be stable but the advantage is not: Competitive implications of key employee mobility. *Strategic Management Journal*, 31, 75–87.
- Alchian, A. A. (1950). Uncertainty, evolution, and economic theory. *Journal of Political Economy*, 58, 211–221.
- Aldrich, H. (1999). *Organizations evolving*. Thousand Oaks, CA: Sage Publications.
- Archer, M. (1995). *Realist social theory: The morphogenetic approach*. Cambridge, UK: Cambridge University Press.
- Argote, L., & Ren, Y. (2012). Transactive memory systems: A microfoundation of dynamic capabilities. *Journal of Management Studies*, 49, 1375–1382.
- Argyres, N., Felin, T., Foss, N., & Zenger, T. (2012). Organizational economics of capability and heterogeneity. *Organization Science*, 23, 1213–1226.
- Arrow, K. J. (1951). *Social choice and individual values*. New Haven, CT: Yale University Press.
- Asch, S. E. (1951). Effects of group pressure upon the modification and distortion of judgments. In H. Guetzkow (Ed.), *Groups, leadership and men: Research in human relations*. Pittsburgh: Carnegie Press.
- Banerjee, A. V. (1992). A simple model of herd behavior. *Quarterly Journal of Economics*, 107, 797–818.
- Bapuji, H., Hora, M., & Saeed, A. M. (2012). Intentions, intermediaries, and interaction: Examining the emergence of routines. *Journal of Management Studies*, 49, 1586–1607.
- Barney, J. B. (1986). Strategic factor markets: Expectations, luck and business strategy. *Management Science*, 32, 1231–1241.
- Barth, R. (2007). Overview: Sixty years in anthropology. *Annual Review of Anthropology*, 36, 1–16.
- Beckman, C. M. (2006). The influence of founding team company affiliations on firm behavior. *Academy of Management Journal*, 49, 741–758.
- Bower, G. H., & Hilgard, E. R. (1981). *Theories of learning*. Englewood Cliffs, NJ: Prentice Hall.
- Caldwell, B. (2005). *Hayek's challenge: An intellectual biography of F. A. Hayek*. Chicago: University of Chicago Press.
- Campbell, B. A., Coff, R., & Kruscynski, D. (2010). Rethinking sustained competitive advantage from hu-

- man capital. *Academy of Management Review*, 37, 376–395.
- Carlile, P. (2002). A pragmatic view of knowledge and boundaries: Boundary objects in new product development. *Organization Science*, 13, 442–455.
- Cederman, L. E. (2005). Computational models of social forms: Advancing generative process theory. *American Journal of Sociology*, 110, 864–893.
- Chatman, J. (1991). Matching people and organizations: Selection and socialization in public accounting firms. *Administrative Science Quarterly*, 36, 459–484.
- Chwe, M. (2001). *Rational ritual: Culture, coordination and common knowledge*. Princeton, NJ: Princeton University Press.
- Coase, R. (1937). The nature of the firm. *Economica*, 4, 386–405.
- Coff, R. (1999). When competitive advantage doesn't lead to performance: Resource-based theory and stakeholder bargaining power. *Organization Science*, 10, 119–133.
- Coff, R., & Kriscynski, D. (2011). Drilling for micro-foundations of human capital-based competitive advantages. *Journal of Management*, 37, 1429–1443.
- Cohen, W., & Levinthal, D. A. (1990). Absorptive capacity: A new perspective on learning and innovation. *Administrative Science Quarterly*, 35, 128–152.
- Coleman, J. (1990). *Foundations of social theory*. Boston: Harvard University Press.
- Dansereau, F., Yammarino, F. J., & Kohles, J. C. (1999). Multiple levels of analysis from a longitudinal perspective: Some implications for theory building. *Academy of Management Review*, 24, 346–357.
- Davis-Blake, A., & Pfeffer, J. (1989). Just a mirage: The search for dispositional effects in organizational research. *Academy of Management Review*, 14, 385–400.
- Denrell, J., Fang, C., & Winter, S. G. (2003). The economics of strategic opportunity. *Strategic Management Journal*, 24, 977–990.
- Diehl, M., & Stroebe, W. (1991). Productivity loss in idea-generating groups: Tracking down the blocking effect. *Journal of Personality and Social Psychology*, 61, 392–403.
- DiMaggio, P. J., & Powell, W. W. (1991). Introduction. In W. W. Powell & P. J. DiMaggio (Eds.), *The new institutionalism in organizational analysis*. Chicago: University of Chicago Press.
- Eisenhardt, K., Furr, N. R., & Bingham, C. B. (2010). Microfoundations of performance: Balancing efficiency and flexibility in dynamic environments. *Organization Science*, 21, 1263–1273.
- Eisenhardt, K., & Martin, J. (2000). Dynamic capabilities: What are they? *Strategic Management Journal*, 21, 1105–1121.
- Elster, J. (1989). *Nuts and bolts for the social sciences*. Cambridge, UK: Cambridge University Press.
- Estes, W. K. (1970). *Learning theory and mental development*. New York: Academic Press.
- Fehr, E., & Tyran, J. R. (2005). Individual rationality and aggregate outcomes. *Journal of Economic Perspectives*, 19, 43–66.
- Felin, T. (2012). Cosmologies of capability, markets and wisdom of crowds: Introduction and comparative agenda. *Managerial and Decision Economics*, 33, 283–294.
- Felin, T., & Foss, N. (2005). Strategic organization: A field in search of micro-foundations. *Strategic Organization*, 3, 441–455.
- Felin, T., & Foss, N. (2011). The endogenous origins of experience, routines and capabilities: The poverty of stimulus. *Journal of Institutional Economics*, 7, 231–256.
- Felin, T., Foss, N., Heimeriks, K., & Madsen, T. (2012). Microfoundations of routines and capabilities: Individuals, processes, and structure. *Journal of Management Studies*, 49, 1351–1374.
- Felin, T., & Hesterly, W. (2007). The knowledge-based view, nested heterogeneity, and new value creation: Philosophical considerations on the locus of knowledge. *Academy of Management Review*, 32, 195–218.
- Felin, T., & Knudsen, T. (2012). A theory of nascent entrepreneurship and organization. *Managerial and Decision Economics*, 33, 409–426.
- Felin, T., & Zenger, T. R. (2011). Information aggregation, matching and radical market-hierarchy hybrids: Implications for the theory of the firm. *Strategic Organization*, 9, 163–173.
- Foss, N. J. (2003). Selective intervention and internal hybrids: Interpreting and learning from the rise and decline of the Oticon spaghetti organization. *Organization Science*, 14, 331–349.
- Foss, N. J., & Lindenberg, S. (2013). Micro-foundations for strategy: A goal-framing perspective on the drivers of value creation. *Academy of Management Perspectives*, 27(2).
- Gary, M. S., Wood, R. E., & Pillinger, T. (2012). Enhancing mental models, analogical transfer, and perfor-

- mance in strategic decision making. *Strategic Management Journal*, 33(11), 1229–1246.
- Gavetti, G. (2005). Cognition and hierarchy: Rethinking the microfoundations of capabilities' development. *Organization Science*, 16, 599–617.
- Gavetti, G. (2012). Toward a behavioral theory of strategy. *Organization Science*, 23(1), 267–285.
- Gavetti, G., Greve, H., Levinthal, D., & Ocasio, W. (2012). The behavioral theory of the firm: Assessment and prospects. *Academy of Management Annals*, 6(1), 1–40.
- Gavetti, G., Levinthal, D., & Ocasio, W. (2007). Neo-carnegie: The Carnegie School's past, present, and re-constructing for the future. *Organization Science*, 18, 523–536.
- Gavetti, G., Levinthal, D. A., & Rivkin, J. W. (2005). Strategy making in novel and complex worlds: The power of analogy. *Strategic Management Journal*, 26, 691–712.
- Giddens, A. (1984). *The constitution of society*. Berkeley: University of California Press.
- Greve, H. (2013). Microfoundations of management: Behavioral strategies and levels of rationality in organizational action. *Academy of Management Perspectives*, 27(2).
- Hambrick, D. C., & Mason, P. A. (1984). Upper echelons: The organization as a reflection of its top managers. *Academy of Management Review*, 9, 193–206.
- Hann, C., & Hart, K. (2011). *Economic anthropology*. Cambridge, UK: Polity Press.
- Harper, D. A., & Lewis, D. A. (2012). New perspectives on emergence in economics. *Journal of Economic Behavior and Organization*, 82(2–3), 329–337.
- Harrison, J. R., & Carroll, G. R. (2005). *Culture and demography in organizations*. Princeton, NJ: Princeton University Press.
- Hayek, F. (1945). The use of knowledge in society. *American Economic Review*, 35, 519–530.
- Heath, C., & Sitkin, S. (2001). Big-B versus big-O: What is organizational about organizational behavior? *Journal of Organizational Behavior*, 2, 43–58.
- Henderson, R., & Cockburn, I. (1994). Measuring competence? Exploring firm effects in pharmaceutical research. *Strategic Management Journal*, 15, 63–84.
- Hirschman, A. (1971). *Exit, voice and loyalty*. Princeton, NJ: Princeton University Press.
- Hodgson, G. (2012). The mirage of microfoundations. *Journal of Management Studies*, 49, 1389–1394.
- Hodgson, G., & Knudsen, T. (2011a). Poverty of stimulus and absence of cause: Some questions for Felin and Foss. *Journal of Institutional Economics*, 7, 295–298.
- Hodgson, G., & Knudsen, T. (2011b). *Darwin's conjecture: The search for general principles of social and economic evolution*. Chicago: University of Chicago Press.
- Holyoak, K. J., & Thagard, P. (1996). *Mental leaps: Analogy in creative thought*. Cambridge, MA: MIT Press.
- House, R. J., Shane, S. A., & Herold, D. M. (1996). Rumors of the death of dispositional research are vastly exaggerated. *Academy of Management Review*, 21, 203–263.
- Ingram, P., & Clay, K. (2000). The choice-within-constraints new institutionalism and implications for sociology. *American Sociological Review*, 26, 525–546.
- Janssen, M. (1993). *Microfoundations: A critical inquiry*. London: Routledge.
- Jepperson, R., & Meyer, J. (2011). Multiple levels of analysis and the limitations of methodological individualisms. *Sociological Theory*, 29, 54–73.
- Jeppesen, L. B., & Lakhani, K. R. (2010). Marginality and problem-solving effectiveness in broadcast search. *Organization Science*, 18, 1016–1033.
- Johnson, V. (2007). What is organizational imprinting? Cultural entrepreneurship in the founding of the Paris opera. *American Journal of Sociology*, 113, 97–113.
- Katz, E., & Lazarsfeld, P. F. (1955). *Personal influence: The part played by people in the flow of mass communications*. Glencoe, IL: Free Press.
- Kim, J. (1993). *Supervenience and mind*. Cambridge, UK: Cambridge University Press.
- King, B., Felin, T., & Whetten, D. (2010). Finding the organization in organization theory: A meta-theory of the organization as social actor. *Organization Science*, 21, 290–305.
- Knudsen, T., & Levinthal, D. (2007). Two faces of search: Alternative generation and alternative evaluation. *Organization Science*, 18, 39–54.
- Kogut, B., & Zander, U. (1992). Knowledge of the firm, combinative capabilities, and the replication of technology. *Organization Science*, 3, 383–397.
- Kozlowski, S., & Chao, G. T. (2012). The dynamics of emergence: Cognition and cohesion in work teams. *Managerial and Decision Economics*, 33(5–6), 335–354.

- Langley, A. (2007). Processing thinking in strategic organization. *Strategic Organization*, 5, 271–282.
- Laureiro-Martinez, D., Brusoni, S., & Zollo, M. (2010). The neuroscientific foundations of the exploration-exploitation dilemma. *Journal of Neurosciences, Psychology and Economics*, 3, 95–115.
- Levallois, C., Clithero, J. A., Wouters, P., Smidts, A., & Huettel, S. A. (2012). Translating upwards: Linking neural and social sciences via neuroeconomics. *Nature Reviews Neuroscience*, 13, 789–797.
- Levinthal, D. A. (2011). A behavioral approach to strategy—What's the alternative? *Strategic Management Journal*, 32, 1517–1523.
- Lewis, K., & Herndon, B. (2011). Transactive memory systems: Current issues and future research directions. *Organization Science*, 22, 1254–1265.
- Macy, M. W., & Willer, R. (2002). From factors to actors: Computational sociology and agent-based modeling. *Annual Review of Sociology*, 28, 143–166.
- Malmgren, H. B. (1961). Information, expectations, and the theory of the firm. *Quarterly Journal of Economics*, 75, 399–421.
- March, J. (1962). The business firm as a political coalition. *Journal of Politics*, 24, 662–678.
- March, J. (1991). Exploration and exploitation in organizational learning. *Organization Science*, 2, 71–87.
- Mayr, E. (1961). Cause and effect in biology. *Science*, 134, 1501–1506.
- McPherson, M., Smith-Lovin, L., & Cook, J. M. (2001). Birds of a feather: Homophily in social networks. *Annual Review of Sociology*, 27, 415–444.
- Molitero, T., & Ployhart, R. E. (2011). Emergence of the human capital resource: A multilevel model. *Academy of Management Review*, 36, 127–150.
- Molloy, J. C., Ployhart, R. E., & Wright, P. M. (2011). The myth of “the” micro-macro divide: Bridging system-level and disciplinary divides. *Journal of Management*, 37, 581–609.
- Mowday, R. T., & Sutton, R. I. (1993). Organizational behavior: Linking individuals and groups to organizational contexts. *Annual Review of Psychology*, 44, 195–229.
- Murmann, J. P., Aldrich, H., Levinthal, D., & Winter, S. (2003). Evolutionary thought in management and organization theory in the beginning of the new millennium. *Journal of Management Inquiry*, 12, 1–19.
- Nagel, E. (1961). *The structure of science: Problems in the logic of scientific explanation*. New York: Harcourt, Brace & World.
- Nelson, R. R., & Winter, S. (1982). *An evolutionary theory of economic change*. Boston: Harvard University Press.
- Nickerson, J. A., & Zenger, T. R. (2004). A knowledge-based theory of the firm: The problem-solving perspective. *Organization Science*, 15, 617–632.
- Nickerson, J. A., & Zenger, T. R. (2008). Envy, comparison costs, and the economic theory of the firm. *Strategic Management Journal*, 29, 1429–1449.
- Oppenheim, P., & Putnam, H. (1958). Unity of science as a working hypothesis. In H. Feigl, M. Scriven, & G. Maxwell (Eds.), *Minnesota studies in the philosophy of science* (pp. 3–36). Minneapolis: University of Minnesota Press.
- Pettigrew, A. (1992). The character and significance of strategy process research. *Strategic Management Journal*, 13, 5–16.
- Ployhart, R. E., & Molitero, T. P. (2011). Emergence of the human capital resource: A multilevel model. *Academy of Management Review*, 36, 127–150.
- Polanyi, K. (1944). *The great transformation: The political and economic origins of our time*. New York: Holt, Reinhardt & Winston.
- Popper, K. (1959). *The logic of scientific discovery*. New York: Hutchinson.
- Porter, D. P., & Smith, V. (2003). Stock market bubbles in the laboratory. *Journal of Behavioral Finance*, 4, 7–20.
- Powell, T. C., Lovullo, D., & Fox, C. R. (2011). Behavioral strategy. *Strategic Management Journal*, 32, 1369–1386.
- Powell, T. C., & Puccinelli, N. M. (2012). The brain as a substitute for strategic organization. *Strategic Organization*, 10, 207–214.
- Raub, W., Buskens, V., & Van Assen, M. (2011). Micro-macro links and microfoundations in sociology. *Journal of Mathematical Sociology*, 35, 1–25.
- Rietzschel, E. F., Nijstad, B. A., & Stroebe, W. (2006). Productivity is not enough: A comparison of interactive and nominal brainstorming groups on idea generation and selection. *Journal of Experimental Social Psychology*, 42, 244–251.
- Rousseau, D. M. (1985). Issues of levels in organizational research: Multi-level and cross-level. *Research in Organizational Behavior*, 7, 1–37.
- Sah, R. K., & Stiglitz, J. E. (1986). The architecture of economic systems: Hierarchies and polyarchies. *American Economic Review*, 76, 716–727.

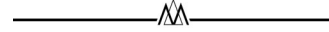
- Sawyer, R. K. (2001). Emergence in sociology: Contemporary philosophy of mind and some implications for sociology. *American Journal of Sociology*, 107, 551–585.
- Sawyer, R. K. (2005). *Social emergence: Societies as complex systems*. Cambridge, UK: Cambridge University Press.
- Schaffer, J. (2003). Is there a fundamental level? *Nous*, 37, 498–517.
- Schelling, T. C. (1971). Dynamic models of segregation. *Journal of Mathematical Sociology*, 1, 143–186.
- Schelling, T. C. (1978). *Micromotives and macrobehavior*. New York: W. W. Norton.
- Schneider, B. (1987). The people make the place. *Personnel Psychology*, 40, 437–454.
- Selznick, P. (1957). *Leadership in administration*. New York: Harper and Row.
- Selznick, P. (1996). Institutionalism old and new. *Administrative Science Quarterly*, 41, 270–277.
- Sen, A. (1999). The possibility of social choice. *American Economic Review*, 89, 349–378.
- Shefir, M., & Hovland, C. I. (1961). *Social judgment: Assimilation and contrast effects in communication and attitude*. New Haven, CT: Yale University Press.
- Shiller, R. J. (2003). From efficient markets theory to behavioral finance. *Journal of Economic Perspectives*, 17, 83–104.
- Simmel, G. (1974). *Individuality and social forms*. Chicago: University of Chicago Press.
- Simon, H. A. (1956). Rational choice and the structure of the environment. *Psychological Review*, 63, 129–138.
- Spender, J. C. (1996). Making knowledge the basis of a dynamic theory of the firm. *Strategic Management Journal*, 17, 45–62.
- Stangor, C. (2009). *Social groups in action and interaction*. Oxford, UK: Psychology Press.
- Stinchcombe, A. L. (1965). Social structure and organizations. In J. G. March (Ed.), *Handbook of organizations* (pp. 142–193). New York: Rand McNally.
- Stinchcombe, A. (1990). *Information and organizations*. Berkeley and Los Angeles: University of California Press.
- Stirling, W. C., & Felin, T. (2013). Game theory, conditional preferences and social influence. *PLoS One*, 8(2), e56751.
- Swedberg, R. (2007). *Principles of economic sociology*. Princeton, NJ: Princeton University Press.
- Udehn, L. (2001). *Methodological individualism: Background, history and meaning*. London: Routledge.
- Udehn, L. (2002). The changing face of methodological individualism. *Annual Review of Sociology*, 28, 479–507.
- Vaugiraud, V. (2005). Beliefs, bailouts and the spread of bank panics. *Bulletin of Economic Research*, 57, 93–107.
- Volk, S., & Kohler, T. (2012). Brains and games: Applying neuroeconomics to organizational research. *Organizational Research Methods*, 15, 522–552.
- Weber, M. (1949). *The methodology of the social sciences*. New York: Free Press.
- Weintraub, E. R. (1979). *Microfoundations: The compatibility of microeconomics and macroeconomics*. Cambridge, UK: Cambridge University Press.
- Whetten, D., Felin, T., & King, B. (2009). The practice of theory borrowing in organizational studies: Current issues and future directions. *Journal of Management*, 35, 537–563.
- Whittington, R. (2003). The work of strategizing and organizing: For a practice perspective. *Strategic Organization*, 1, 117–125.
- Winter, S. G. (2011). Problems at the foundation? Comments on Felin and Foss. *Journal of Institutional Economics*, 7, 257–277.
- Winter, S. G. (2012a). Capabilities: Their origins and ancestry. *Journal of Management Studies*, 49, 1402–1406.
- Winter, S. G. (2012b). Purpose and progress in the theory of strategy: Comments on Gavetti. *Organization Science*, 23, 1213–1226.
- Winter, S. G. (2013). Habit, deliberation and action: Strengthening the microfoundations of routines and capabilities. *Academy of Management Perspectives*, 27(2).
- Zollo, M., & Winter, S. G. (2002). Deliberate learning and the evolution of dynamic capabilities. *Organization Science*, 13, 339–352.



Jay B. Barney (jay.barney@business.utah.edu) is a Presidential Professor and Lasonde Chair in Social Entrepreneurship at the Eccles School of Business at the University of Utah. Professor Barney's research in strategic management has focused on the relationship between a firm's resources and capabilities and its performance. His research in entrepreneurship has focused on the processes through which individuals create the entrepreneurial opportunities they exploit.

Teppo Felin (teppo.felin@byu.edu) is an Associate Professor and Lee Perry Fellow at the Marriott School at BYU. Starting in August 2013 he will be a Professor at Saïd Business School at the University of Oxford. Professor Felin's research focuses on the microfoundations of strategy and organization, the origins of capability, complex systems, and organizational forms. He is a co-editor

of the journal *Strategic Organization*. His research has been published in *Academy of Management Review*, *Organization Science*, *Erkenntnis*, *PLOS ONE*, *Journal of Management*, *Journal of Management Studies*, and numerous other outlets.



Copyright of Academy of Management Perspectives is the property of Academy of Management and its content may not be copied or emailed to multiple sites or posted to a listserv without the copyright holder's express written permission. However, users may print, download, or email articles for individual use.